

Lecture Notes

Course: Complex Analysis

TA: Morales C. Ronny

Topic: The Exponential Function

Theorem. There exists a function $E : \mathbb{R} \rightarrow \mathbb{R}$ such that:

- i) $E'(x) = E(x)$ for all $x \in \mathbb{R}$.
- ii) $E(0) = 1$.

Proof. Inductively define the sequence (E_n) of (continuous) functions as

$$E_1(x) := 1 + x, \quad E_{n+1}(x) := 1 + \int_0^x d\xi E_n(\xi),$$

we recall that if $f \in C[a, b]$, then $f \in \mathcal{R}[a, b]$ so that $E_n(x)$ is a well-defined and continuous function for all $n \in \mathbb{N}$. Moreover, by the Fundamental Theorem (Second Form) it also follows that

$$E'_{n+1}(x) = E_n(x), \quad \text{for } n \in \mathbb{N}.$$

It is not hard to see that

$$E_n(x) = \sum_{k=0}^n \frac{x^k}{k!}.$$

Now we let $A > 0$ be given; then if $|x| \leq A$ and $m > n > 2A$, we have

$$\begin{aligned} |E_m(x) - E_n(x)| &= \left| \sum_{j=n+1}^m \frac{x^j}{j!} \right| = \left| \frac{x^{n+1}}{(n+1)!} \left(\sum_{j=n+1}^m \frac{(n+1)!}{j!} x^{j-(n+1)} \right) \right| \\ &= \left| \frac{x^{n+1}}{(n+1)!} \right| \left| \sum_{j=n+1}^m \frac{(n+1)!}{j!} x^{j-(n+1)} \right| = \left| \frac{x^{n+1}}{(n+1)!} \right| \left| \sum_{k=1}^{m-n} \frac{(n+1)!}{(n+k)!} x^{k-1} \right|, \end{aligned}$$

but since $1 \leq k \leq m - n$ we have that

$$\frac{(n+1)!}{(n+k)!} = \frac{1}{\prod_{\ell=0}^{k-1} (n+k-\ell)} \leq \frac{1}{\prod_{\ell=0}^{k-1} n} = \frac{1}{n^k} \leq \frac{1}{n^{k-1}}$$

the inequality following from $n+k-\ell \geq n$ (since $\ell \leq k-1$) which implies $1/(n+k-\ell) \leq 1/n$ so that

$$\begin{aligned} |E_m(x) - E_n(x)| &= \left| \frac{x^{n+1}}{(n+1)!} \right| \left| \sum_{k=1}^{m-n} \frac{(n+1)!}{(n+k)!} x^{k-1} \right| \\ &\leq \left| \frac{x^{n+1}}{(n+1)!} \right| \left| \sum_{k=1}^{m-n} \frac{1}{n^{k-1}} x^{k-1} \right| \leq \left| \frac{A^{n+1}}{(n+1)!} \right| \left| \sum_{k=1}^{m-n} \frac{A^{k-1}}{n^{k-1}} \right| < \left| \frac{A^{n+1}}{(n+1)!} \right| \sum_{k=0}^{\infty} \left(\frac{A}{n} \right)^k = \left| \frac{A^{n+1}}{(n+1)!} \right| \frac{1}{1 - A/n}, \end{aligned}$$

the last inequality following from the fact that $A > 0$. But since $\lim(A^n/n!) = 0$ (by the ratio test), we have that given ε , we can choose H such that if $n > H$, then $A^{n+1}/(n+1)! < \varepsilon/2$ and thus for $m > n > \sup\{H, 2A\}$ we get

$$|E_m(x) - E_n(x)| < \varepsilon.$$

We therefore conclude that the sequence (E_n) is Cauchy (and thus convergent), independently of x so that (E_n) converges uniformly on $[-A, A]$, and since A is arbitrary, (E_n) converges uniformly on $\mathbb{R}$.

We now let $E_n \rightarrow E$, this way E is continuous by uniform convergence and continuity of each E_n , but since we have $E'_{n+1} = E_n$, the sequence of derivatives (E'_n) is also uniformly convergent and thus $E'_n \rightarrow E'$ but notice that

$$E'(x) = \lim(E'_n(x)) = \lim(E_{n-1}(x)) = \lim(E_n(x)) = E(x) \quad \text{for } x \in \mathbb{R}$$

and hence (i) has been proven, for (ii) notice that $E_n(0) = 1$ for all $n \in \mathbb{N}$ and thus $E(0) = 1$.

Lemma. The unique function $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $F'(x) = F(x)$ and $F(0) = 0$ is the zero function given by $F(x) = 0$ for all $x \in \mathbb{R}$.

Proof. Suppose $F' = F$ and $F(0) = 0$ (this clearly implies that $F^{(k)}(0) = 0$), this way, let $I_x := [0, x]$ so that by Taylor's theorem, if $x_0 \in I_x$, then there exists c between x and 0 such that

$$F(x) = \sum_{k=0}^{n-1} \frac{F^{(k)}(0)}{k!} x^k + \frac{F^{(n)}(c)}{n!} x^n,$$

but by continuity of F we have that there exists $M > 0$ such that $|F(t)| \leq M$ whenever $t \in I_x$, then (since $c \in I_x$)

$$F(x) = \frac{F^{(n)}(c)}{n!} x^n = \frac{F(c)}{n!} x^n \implies |F(x)| \leq \frac{M}{n!} |x|^n,$$

and again we have that $\lim(|x|^n/n!) = 0$ for any $x \in \mathbb{R}$ so that $F(x) = 0$ for $x \in \mathbb{R}$.

Corollary. The function $E : \mathbb{R} \rightarrow \mathbb{R}$ that satisfies (i) and (ii) of the previous theorem is unique.

Proof. Take $F(x) = E_1(x) - E_2(x)$ with E_1, E_2 two functions satisfying i) and ii)

Theorem. The exponential function satisfies the following properties:

- (iii) $E(x) \neq 0$ for all $x \in \mathbb{R}$;
- (iv) $E(x+y) = E(x)E(y)$ for all $x, y \in \mathbb{R}$;
- (v) $E(rx) = (E(x))^r$ for all $r \in \mathbb{Q}$.

Proof. (i) Suppose that $E(\alpha) = 0$ for $\alpha \in \mathbb{R}$, then choose $I_\alpha := [0, \alpha]$ and let K be such that $|E(t)| \leq K$ for $t \in I_\alpha$, by Taylor's theorem (choosing $x = 0, x_0 = \alpha$) we have that there exists c between 0 and α for which (notice that $c \in I_\alpha$)

$$E(0) = \sum_{\ell=0}^{n-1} \frac{E^{(\ell)}(\alpha)}{\ell!} (-\alpha)^\ell + \frac{E^{(n)}(c)}{n!} (-\alpha)^n = \frac{E^{(n)}(c)}{n!} (-\alpha)^n = \frac{E(c)}{n!} (-\alpha)^n$$

but this would imply (same as before) that $E(0) = 0$, a contradiction.

(iv) Define $G(x) := E(x+y)/E(y)$, and since G clearly satisfies (i) and (ii), by the uniqueness theorem we have that $G(x) = E(x)$.

(v) Similarly, one can define $H(x) := E(rx) - (E(x))^r$ so that applying the chain rule, one gets that $H(0) = 0$ and $H'(x) = H(x)$ so that a similar argument as before yields the desired result ($H(x) \equiv 0$).

Example. Compute the modulus of $\exp z$.

Given that $\exp z = \lim s_n(z)$ for any $z \in \mathbb{C}$ with

$$s_n(z) = \sum_{k=0}^n \frac{z^k}{k!} \implies s_n^*(z) = \sum_{k=0}^n \frac{(z^*)^k}{k!} = s_n(z^*),$$

we have that $\lim s_n^*(z) = (\lim s_n(z))^* = (\exp z)^*$ and $\lim s_n^*(z) = \lim s_n(z^*) = \exp(z^*)$. We therefore have

$$|\exp z|^2 = \exp z \exp z^* = \exp(z + z^*) = \exp 2\Re(z) = (\exp \Re(z))^2 \iff |\exp z| = \exp \Re(z).$$

Theorem 7.14. Given any $z \in \mathbb{C} \setminus \{0\}$, there are unique $r > 0$ and $w \in \mathbb{C}$ with $|w| = 1$ such that $z = rw$

Proof. For any $z \neq 0$ write $z = |z|z/|z|$, so that $r = |z|$ and $w = z/|z|$ have the required properties. For uniqueness, if $z = r_1w_1 = r_2w_2$ with $r_1, r_2 > 0$ and $|w_1| = |w_2| = 1$ so $|r_1w_1| = |r_2w_2|$ implies $r_1 = r_2$ and this also implies $w_2 = w_1$.

Theorem 7.15. Let w be a complex number with unit modulus. Then there exists $\theta \in \mathbb{R}$ such that $-\pi < \theta \leq \pi$ and $w = \exp(i\theta)$.

Proof. Let $w = u + iv$ with $u, v \in \mathbb{R}$ and $u^2 + v^2 = 1$.

- Case *i*) Suppose $0 < u < 1$, this condition implies $-1 < v < 1$ and $v \neq 0$, first suppose $0 < v < 1$. Since $\cos$ is a continuous function and strictly decreasing on $[0, \pi/2]$ and $\cos 0 = 1, \cos \pi/2 = 0$, we have $\cos \pi/2 < u < \cos 0$, then there exists a unique $\theta \in (0, \pi/2)$ for which $\cos \theta = u$ but then $v^2 = 1 - \cos^2 \theta = \sin^2 \theta$ but since $\theta \in (0, \pi/2)$, then $\sin \theta > 0$ so that $v = \sin \theta$ because $v > 0$. Hence $w = \cos \theta + i \sin \theta = \exp(i\theta)$.
- Case *ii*) Suppose again $0 < u < 1$ but now with $-1 < v < 0$ then $u + iv = -i(-v + iu) = -i \exp(i\theta) = \exp(i(\theta - \pi/2))$ with $\theta \in (0, \pi/2)$ by the same argument as before.
- Case *iii*) Suppose $-1 < u < 0$, this implies again that $-1 < v < 1$ with $v \neq 0$, so begin by assuming that $0 < v < 1$, then $u + iv = i(-iu + v) = i \exp(i\theta) = \exp(i\pi/2) \exp(i\theta) = \exp(i(\theta + \pi/2))$.
- Case *iv*) Suppose again $-1 < u < 0$ but now with $-1 < v < 0$, then $u + iv = -(-u - iv) = -\exp(i\theta) = \exp(i(\theta - \pi))$ with $\theta \in (0, \pi/2)$.
- Case *v*) Suppose now that either $u = 1$ or $u = -1$, then (in either case) $v = 0$, if $u = 1$ then $\theta = 0$, and if $u = -1$ then $\theta = \pi$.
- Case *vi*) Suppose $u = 0$, then either $v = 1$ or $v = -1$, if $v = 1$ we get $\theta = \pi/2$ and if $v = -1$ we get $\theta = -\pi/2$.

We therefore conclude that for writing $w = \exp(i\phi)$ there is a unique ϕ satisfying

$$\phi \in \begin{cases} (0, \pi/2) & \text{Case } i) \\ (-\pi/2, 0) & \text{Case } ii) \\ (\pi/2, \pi) & \text{Case } iii) \\ (-\pi, \pi/2) & \text{Case } iv) \\ \{0, \pi\} & \text{Case } v) \\ \{-\pi/2, \pi/2\} & \text{Case } vi) \end{cases}$$

so that $\pi < \phi \leq \pi$, as required.

Theorem 7.16. $\exp(z) = 1$ if and only if $z = 2k\pi i$ for some $k \in \mathbb{Z}$.

Proof. $(\Leftarrow)$ Trivial. $(\Rightarrow)$ The solution to $\cos \theta = 1$ and $\sin \theta = 0$ is given by all the $\theta = 2k\pi$ with $k \in \mathbb{Z}$ from where the conclusion readily follows.

Theorem 7.17 (De Moivre's Formula). The conclusion is immediate from writing z in its polar representation

$$z^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta).$$

Theorem 7.18. By stroke of insight (using Theorem 7.16) we see that the numbers

$$\left\{ \exp \frac{2k\pi i}{n} \right\}_{k \in \mathbb{Z}}$$

are all n th roots of unity and lie on the unit circle centred at the origin and are equally spaced around the circle every $2\pi/n$ radians. Hence, all the distinct n n th roots of unity are the numbers

$$\left\{ \exp \frac{2k\pi i}{n} \right\}_{k=0}^{n-1} = \{\zeta^k\}_{k=0}^{n-1}, \quad \zeta = \exp \frac{2\pi i}{n}.$$

Theorem 7.19. We look for the numbers w satisfying $w^n = z$, we write $w = Re^{i\phi}$ and $z = re^{i\theta}$, this way, we get the requirement

$$R^n e^{in\phi} = re^{i\theta},$$

whence $|R^n e^{in\phi}| = |re^{i\theta}|$ so that $|R^n| = |r|$ but since $R \geq 0$ and $r \geq 0$, we get that $R^n = r$ and there is a unique (real) solution to this, in particular, $R = \sqrt[n]{r}$, and then the previous equation reduces to

$$e^{in\phi} = e^{i\theta} \iff e^{i(n\phi - \theta)} = 1 \iff i(n\phi - \theta) = 2k\pi i \iff n\phi - \theta = 2k\pi,$$

so that $\phi = \theta/n + 2k\pi/n$. This way, we get that if t is the unique real positive number for which $t^n = |z| = r$ and if $\theta = \text{Arg } z$, then any number of the form

$$te^{i(\theta/n + 2k\pi/n)} = te^{i\theta/n} e^{2k\pi i/n} \equiv te^{i\theta/n} \zeta^k,$$

is an n th root of z . The set $z^{1/n}$ of all n th roots of z is just

$$z^{1/n} = \{te^{i\theta/n} \zeta^k\}_{k=0}^{n-1},$$

as claimed.